

Final Exam
ODE (MA1150)

Full Marks: 50

Time- 2 hours

Please answer all the questions:

1. If $\phi(x)$ and $\psi(x)$ are two solutions of the differential equation

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = 3f(x), \text{ then find the differential equation of which}$$

$\phi(x) + \psi(x)$ is a solution. [2 marks]

2. Whether two functions x^3 and $x^2|x|$ are linearly independent or not on the interval $(-\infty, 0]$. Justify your answer. [2 marks]

3. Solve $(2xy + e^x)y dx - e^x dy = 0$ to find the general solution. [5 marks]

4. Find the general solution of the given ODE [5 marks]

$$\left(y + \frac{y^3}{3} + \frac{x^2}{2}\right) dx + \frac{1}{4}(x + xy^2) dy = 0.$$

5. Find the integrating factor of the differential equation [3 marks]

$$(\cos y \sin 2x) dx + (\cos^2 y - \cos^2 x) dy = 0 \text{ if exists.}$$

6. Reduce the differential equation $y = 2x \frac{dy}{dx} - y \left(\frac{dy}{dx}\right)^2$ to Clairaut's form by the substitution $y^2 = Y$, $x = X$ and then obtain its general and singular solutions if any. [5 marks]

7. Let $e^{-x} \cos(\sqrt{3}x)$, $e^{-x} \sin(\sqrt{3}x)$, and e^{2x} be three linearly independent solutions of the linear ODE $L[y(x)] = 0$. Then find the general solution of the $L[y(x)] = e^{2x}x^2$. [5 marks]

8. Find the general solution of linear ODE

$$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = e^{e^x} \quad [5 \text{ marks}]$$

9. Find the solution of the ODE

$$y'' - 3y' + 2y = \cos(e^{-x}) \text{ by using the method of variation of parameter.} \quad [5 \text{ marks}]$$

10. If $\phi_1(x)$ is a particular integral of

$$L[y(x)] = \frac{d^2y}{dx^2} - a\frac{dy}{dx} + by = e^{ax} + f(x),$$

and $\phi_2(x)$ is a particular integral of $L[y(x)] = e^{ax} - f(x)$, a, b being constants, then find the particular integral of $L[y(x)] = 2be^{ax}$. [5 marks]

11. If $y_1(x)$ and $y_2(x)$ are two solutions of the differential equation

$$y'' + \tan x y' - \sec x (1 + e^{-x^2}) y = 0 \text{ for } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

with $y_1(0) = 2, y_1'(0) = -1, y_2(0) = 2$ and the Wronskian of $y_1(x)$ and $y_2(x)$ at $x = \pi/4$ is $3\sqrt{2}$, find the value of y_2'' at $x = 0$. [3 marks]

12. Find the solution of the given problem

$$4x^2 \frac{d^2y}{dx^2} + 8x \frac{dy}{dx} + y = 4/\sqrt{x}, \quad y(1) = 0, \quad y(e) = 0. \quad [5 \text{ marks}]$$

Some important formulas

1. If $Mdx + Ndy = 0$ is not exact but M and N are both homogeneous functions of x and y of same degree n , then the integrating factor is $\frac{1}{Mx + Ny}$, provided $Mx + Ny \neq 0$. If $Mx + Ny = 0$, then $\frac{1}{xy}$ or $\frac{1}{x^2}$ or $\frac{1}{y^2}$ or $\frac{1}{x^2 + y^2}$ is an integrating factor.
2. If $\frac{1}{N} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right]$ is a function of x alone, $f(x)$ (say), then $e^{\int f(x) dx}$ is an integrating factor.
3. If $\frac{1}{M} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right]$ is a function of y alone, $\phi(y)$ (say), then $e^{-\int \phi(y) dy}$ is an integrating factor.
4. $\frac{1}{Mx - Ny}$ is an integrating factor, when $M(x, y) = y f_1(xy)$ and $N(x, y) = x f_2(xy)$, provided $Mx - Ny \neq 0$.
5. $x^h y^k$ is an integrating factor of the differential equation $x^\alpha y^\beta (m y dx + n x dy) + x^{\alpha_1} y^{\beta_1} (m_1 y dx + n_1 x dy) = 0$, with $(m n_1 - m_1 n) \neq 0$, where h and k are given by $\frac{h+1+\alpha}{m} = \frac{k+1+\beta}{n}$ and $\frac{h+1+\alpha_1}{m_1} = \frac{k+1+\beta_1}{n_1}$.
6. Consider the ODE $a_0(x) y^{(n)} + a_1(x) y^{(n-1)} + \dots + a_n(x) y = 0, x \in I$, where $a_0(x) \neq 0$. Then the Wronskian $W(y_1, y_2, \dots, y_n; x) = W(y_1, y_2, \dots, y_n; x_0) e^{-\int_{x_0}^x \frac{a_1(x)}{a_0(x)} dx}, x_0 \in I$, for n linearly independent solutions $y_1(x), y_2(x), \dots, y_n(x)$.
7. $\frac{1}{D-\alpha} x = e^{\alpha x} \int x e^{-\alpha x} dx$
8. $\frac{1}{f(D)} e^{ax} \cdot v(x) = e^{ax} \frac{1}{f(D+a)} \cdot v(x)$

*****Best of Luck*****